

From Quantum Many-Body Scars to Slow Dynamics

Exact Diagonalization of a Deformed PXP Chain

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Abstract

Quantum many-body scars and quantum Griffiths regimes provide two distinct routes to slow or incomplete thermalization. Scars are atypical eigenstates embedded in an otherwise thermal spectrum of a clean constrained model, whereas conventional Griffiths behavior is attributed to rare spatial regions in disordered systems. We investigate whether these mechanisms can be connected in a clean deformation of the PXP chain that moves strong $\mathbb{Z}_2$ spectral weight toward low energy. Exact diagonalization identifies low-energy eigenstates with anomalously large $\mathbb{Z}_2$ overlap. Symmetry-resolved spectra determine the canonical energy and heat capacity, while a thermal-quench calculation yields a connected local X -autocorrelation whose decay is consistent with a power law over the selected time window. These finite-size results are compatible with slow Griffiths-like dynamics but do not establish a Griffiths phase. We therefore formulate further tests based on competing dynamical fits, low-temperature heat-capacity scaling, separation between average and typical observables, gap distributions, and persistence over a finite parameter window.

1 Introduction

After exact symmetries and conserved quantities have been resolved, generic interacting quantum systems are expected to thermalize and exhibit quantum-chaotic spectral statistics. Their finite-energy-density eigenstates obey the eigenstate thermalization hypothesis and typically carry volume-law entanglement. Two important exceptions motivate this work. In strongly disordered spin chains and superconducting systems, a quantum Griffiths regime can develop when rare spatial regions generate a broad distribution of relaxation times and exceptionally slow dynamics. In clean Rydberg-blockaded systems such as the PXP chain, quantum many-body scars instead form a sparse set of eigenstates with anomalously low entanglement and strong violations of eigenstate thermalization relative to nearby states.

Both phenomena can retain memory of special initial conditions and produce slow or reviving dynamics, but their usual microscopic explanations are different. Griffiths behavior relies on spatial disorder and rare regions, whereas scarred eigenstates are embedded in the finite-energy-density spectrum of a clean constrained Hamiltonian. Here we investigate a possible connection between them. A deterministic staggered deformation of the PXP model shifts the state with the strongest $\mathbb{Z}_2$ overlap into the low-energy spectrum, where it can influence low-temperature and long-time observables. We combine exact diagonalization, canonical thermodynamics, and a local autocorrelation calculation to test for Griffiths-compatible behavior without disorder. The present finite-size results identify a candidate route rather than a complete demonstration of a disorder-free Griffiths phase.

2 Constrained spin model

Consider a periodic chain of L spin-1/2 degrees of freedom. Let X_i and Z_i denote Pauli operators, and define

$$n_i = \frac{1 + Z_i}{2}, \quad P_i = 1 - n_i = \frac{1 - Z_i}{2}. \quad (1)$$

The Rydberg blockade restricts the Hilbert space to configurations satisfying

$$n_i n_{i+1} = 0 \quad \text{for every } i, \quad (2)$$

with site labels understood modulo L . Within this constrained space, a local spin flip is equivalent to

$$\tilde{X}_i = P_{i-1} X_i P_{i+1}, \quad (3)$$

and the standard PXP Hamiltonian is

$$H_{\text{PXP}} = \sum_{i=0}^{L-1} \tilde{X}_i. \quad (4)$$

The model studied here adds staggered diagonal perturbations,

$$H(g, r) = H_{\text{PXP}} + g \sum_{i=0}^{L-1} (-1)^i (Z_i + r Z_{i-1} Z_i Z_{i+1}). \quad (5)$$

The parameter g sets the overall strength of the deformation and r fixes the relative weight of the three-spin term. The staggering reduces one-site translation symmetry to translation by two sites. Equation (5) is implemented in a custom constrained basis and diagonalized with QuSpin. The deformation is deterministic and introduces no random couplings. Its role is to tune the spectral position and stability of the scarred subspace while preserving a clean setting.

The reference product state is

$$|\mathbb{Z}_2\rangle = |1010 \dots\rangle, \quad (6)$$

which is invariant under translation by two sites and therefore belongs to the zero-momentum sector of that symmetry. Eigenstates with large $\mathbb{Z}_2$ overlap define the scarred subspace used below. Bringing this spectral weight toward the bottom of the spectrum makes the same subspace accessible to low-temperature diagnostics.

3 Exact diagonalization and scarred eigenstates

For the spectral calculation, the Hamiltonian is represented in the two-site translation sector $k = 0$. The current calculation uses $L = 20$ and obtains the full dense eigendecomposition

$$H(g, r) |E_n\rangle = E_n |E_n\rangle. \quad (7)$$

For each eigenstate, the principal scar diagnostic is its overlap with the ordered state,

$$w_n = |\langle \mathbb{Z}_2 | E_n \rangle|^2. \quad (8)$$

The spectral location of the initial state can also be summarized by

$$E_{\mathbb{Z}_2} = \langle \mathbb{Z}_2 | H | \mathbb{Z}_2 \rangle = \sum_n E_n w_n. \quad (9)$$

The numerical spectrum contains a small set of eigenstates whose values of w_n are anomalously large compared with nearby eigenstates. These outliers are quantum many-body scar candidates. For the deformation of principal interest, $g = -0.4$ and $r = 0.2$, the eigenstate with the largest $\mathbb{Z}_2$ overlap lies in the low-energy region of the finite-size spectrum. This is the first ingredient in the proposed connection: a selected nonthermal sector is placed where it can affect low-temperature dynamics. It is not yet a real-space rare region. A central question is whether the scarred sector acts instead as a rare structure in Hilbert or phase space.

The half-chain entanglement entropy of every eigenstate is evaluated after lifting the states from the symmetry-reduced basis to the full constrained basis. It provides an independent diagnostic of whether large-overlap states are atypical relative to their spectral neighbors. Unfolded nearest-neighbor level spacings are computed from the middle two thirds of the spectrum, where edge effects are weaker. Together, the overlap, entanglement, and level statistics distinguish a sparse scarred subspace from a global loss of ergodicity or an accidentally integrable spectrum. A systematic finite-size study is still required for a thermodynamic-limit conclusion.

Figure 1 collects the spectral diagnostics and the corresponding quench dynamics. The upper-left panel displays the anomalously large $\mathbb{Z}_2$ overlaps and places the strongest-overlap state toward the low-energy side of the spectrum. The lower panels show that the same initial state produces long-lived, damped oscillations in a local observable and in the return probability. This spectral selectivity and dynamical memory motivate the thermodynamic and correlation tests that follow.

4 Canonical energy and heat capacity

The thermodynamic calculation uses the eigenvalues from all two-site momentum sectors, rather than only the $k = 0$ sector used for the scar plot. If $E_{n,k}$ denotes the spectrum in an independent momentum block, the partition function is reconstructed as

$$Z(\beta) = \sum_{k \in \mathcal{K}_{\text{ind}}} m_k \sum_n e^{-\beta E_{n,k}}, \quad (10)$$

where $m_k = 2$ for a conjugate pair k and $-k$, while self-conjugate sectors have $m_k = 1$. In practice, the global ground-state energy is subtracted inside the Boltzmann factors to avoid numerical underflow; this shift is restored in the mean energy.

With $k_B = 1$, the canonical internal energy and heat capacity are

$$U(T) = \langle H \rangle_\beta = \frac{1}{Z} \sum_{k,n} m_k E_{n,k} e^{-\beta E_{n,k}}, \quad (11)$$

$$C_V(T) = \frac{\partial U}{\partial T} = \beta^2 \left(\langle H^2 \rangle_\beta - \langle H \rangle_\beta^2 \right). \quad (12)$$

The temperature scan in Figure 2 uses $L = 20$, $r = 0.2$, $0.02 \leq T \leq 2$, and $g = 0, -0.1, \dots, -0.5$. The internal energy rises smoothly with temperature. Each heat-capacity curve has a broad maximum, whose position shifts to higher temperature as the deformation becomes stronger. The scan therefore maps the thermodynamic crossover produced by the same deformation that moves the scarred spectral weight. At finite size, the broad maximum locates the temperature range with the strongest energy fluctuations; it is not by itself a signature of Griffiths behavior or a thermodynamic phase transition.

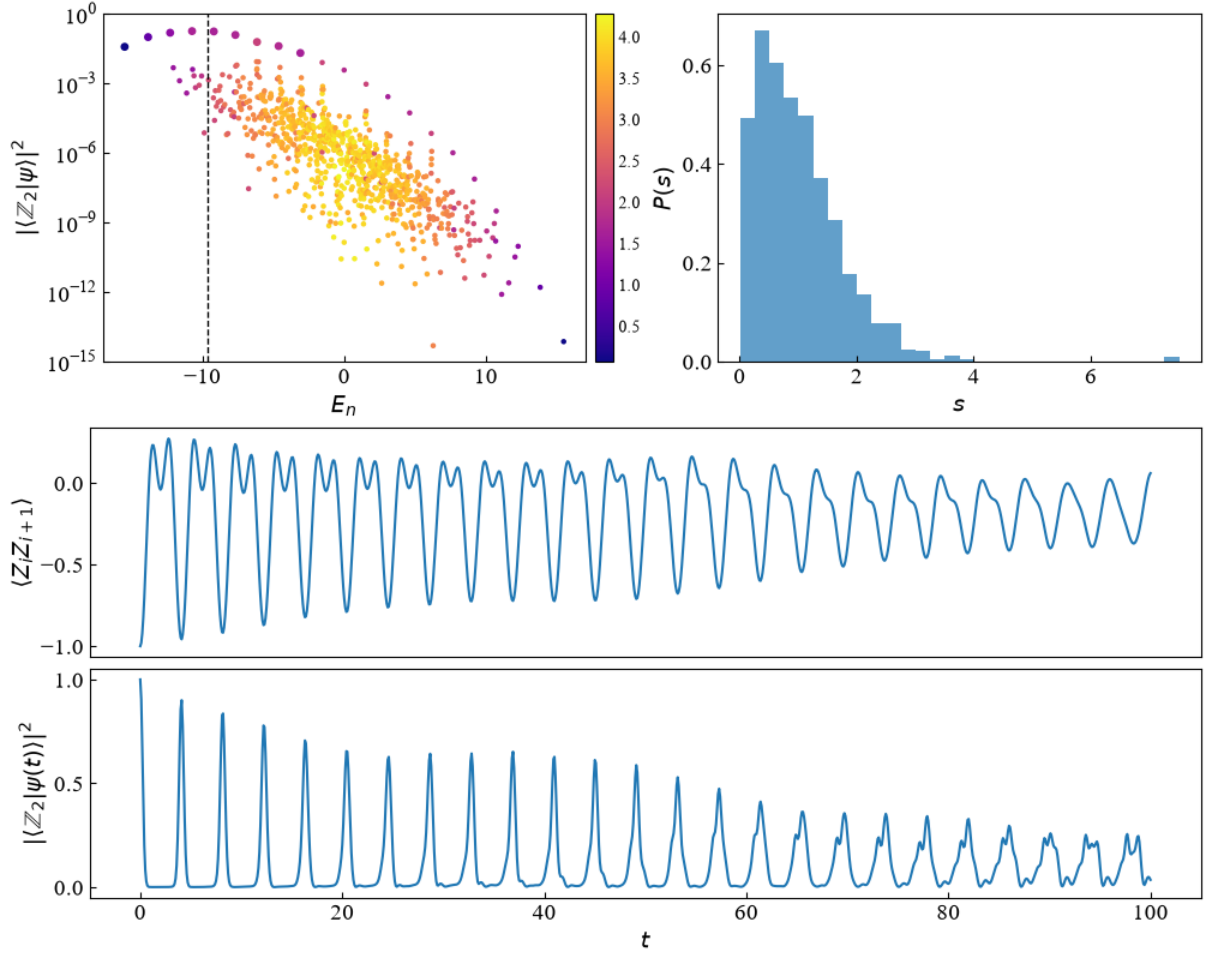


Figure 1: Spectral and dynamical diagnostics for $L = 20$, $g = -0.4$, and $r = 0.2$. Upper left: overlap $|\langle \mathbb{Z}_2 | E_n \rangle|^2$ versus energy; the color scale gives the half-chain entanglement entropy, and the dashed line marks $E_{\mathbb{Z}_2}$. Upper right: distribution of unfolded nearest-neighbor level spacings. Lower panels: the site-averaged nearest-neighbor correlation $L^{-1} \sum_i \langle Z_i Z_{i+1} \rangle$ and the return probability $|\langle \mathbb{Z}_2 | \psi(t) \rangle|^2$ after a quench from $|\mathbb{Z}_2\rangle$.

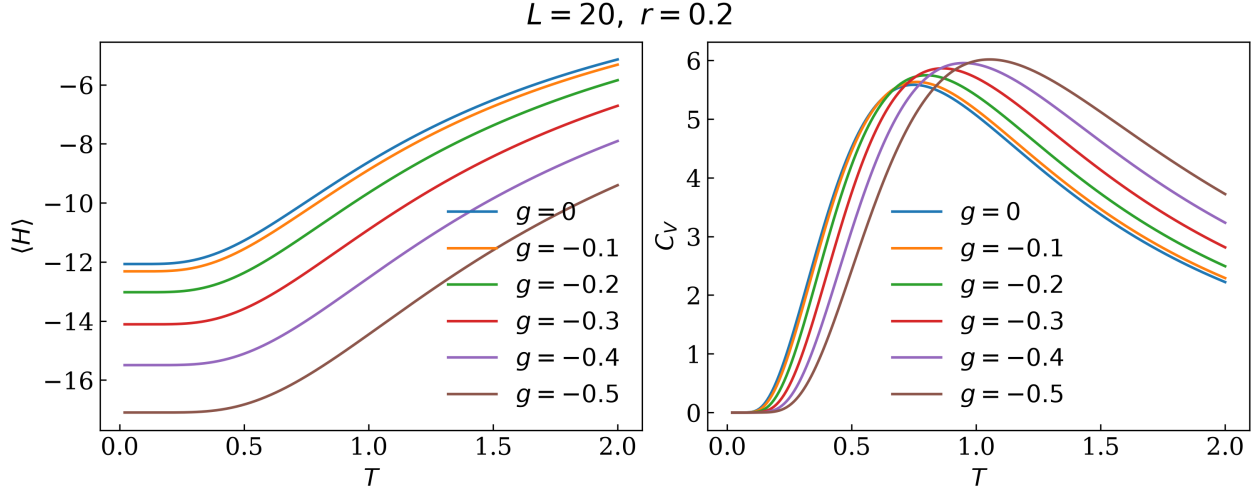


Figure 2: Canonical thermodynamics as a function of temperature for $L = 20$ and $r = 0.2$. The left panel shows the total internal energy $U = \langle H \rangle$; the right panel shows the total heat capacity C_V . Each curve corresponds to a different deformation strength g .

Figure 3 tests the proposed low-temperature form $C_V(T) = AT^\lambda$. For each g , the fit includes temperatures from $T = 0.02$ to one half of the temperature at which C_V reaches its maximum. The fitted curves do not follow the numerical data closely: the coefficients of determination range from $R^2 = 0.845$ to 0.865 , and the fitted exponents depend strongly on g . A finite chain has a discrete low-energy spectrum and generally a nonzero gap Δ . Its asymptotic low-temperature heat capacity is therefore activated,

$$C_V(T) \sim \left(\frac{\Delta}{T}\right)^2 e^{-\Delta/T}, \quad (13)$$

rather than a power law. At higher temperatures, additional levels contribute as the system approaches the broad heat-capacity maximum. The selected interval thus crosses a finite-size crossover instead of isolating a scale-invariant regime. The reported values of λ are window-dependent effective slopes and do not establish Griffiths scaling.

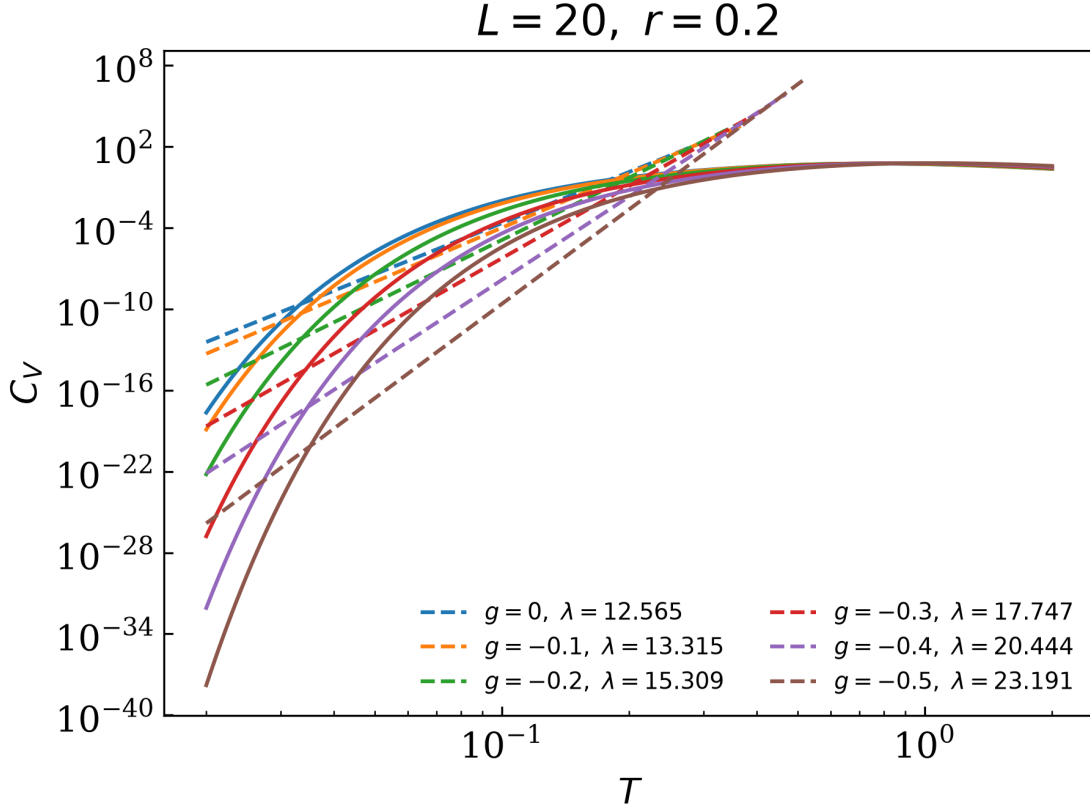


Figure 3: Log-log comparison of the canonical heat capacity (solid curves) and fits to $C_V = AT^\lambda$ (dashed lines) for $L = 20$ and $r = 0.2$. Each fit ends at one half of the corresponding peak temperature. The legend gives the fitted exponent for each deformation strength.

Figure 4 presents the complementary scan in which temperature is fixed and g varies from -0.5 to 0 . At the lowest temperatures, the internal energy follows the deformation of the low-energy spectrum while C_V remains close to zero. At higher temperatures, the heat capacity becomes sensitive to g across the full interval. Taken together, Figures 2 and 4 show how the same spectral rearrangement appears along two cuts through the (g, T) plane. They also establish a baseline for a sharper test proposed in this work: a Griffiths regime should produce a non-universal low-temperature law $C_V(T) \sim T^\lambda$ over the same finite interval of g in which slow dynamics is observed. Testing that statement requires low-temperature log-log fits, several system sizes, and control of the finite-size gap.

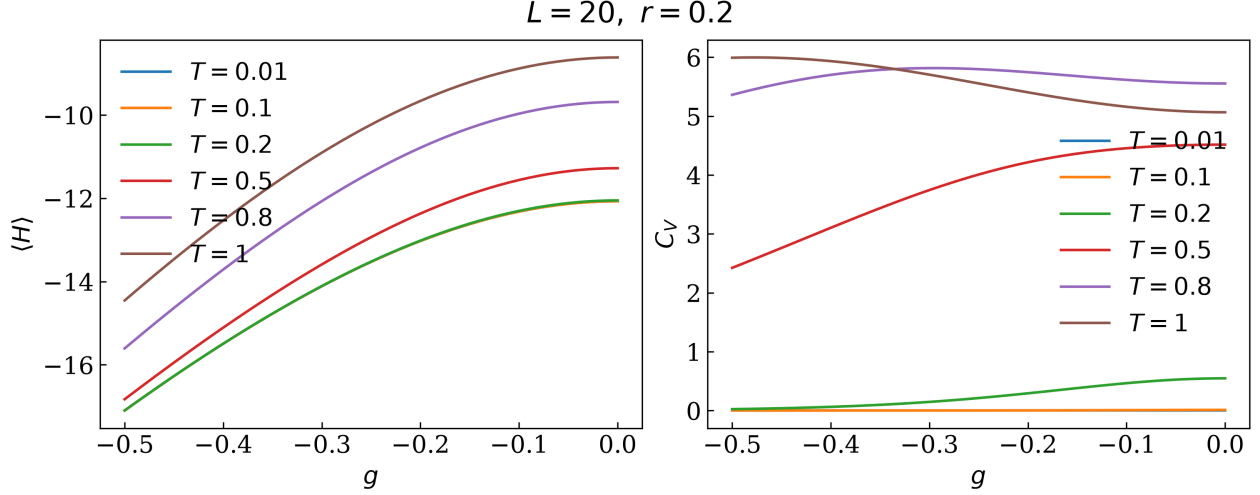


Figure 4: Canonical thermodynamics as a function of deformation strength for $L = 20$ and $r = 0.2$. The left and right panels show $U(g)$ and $C_V(g)$, respectively, at fixed temperatures $T = 0.01, 0.1, 0.2, 0.5, 0.8$, and 1 .

5 Connected local autocorrelation

The dynamical calculation starts from a canonical state of the undeformed PXP Hamiltonian,

$$\rho_0 = \frac{e^{-\beta H_{\text{PXP}}}}{\text{Tr}(e^{-\beta H_{\text{PXP}}})}, \quad (14)$$

and evolves operators with the deformed Hamiltonian $H_1 = H(g, r)$,

$$\tilde{X}_i(t) = e^{iH_1 t} \tilde{X}_i e^{-iH_1 t}. \quad (15)$$

The measured quantity is the site-averaged connected autocorrelation

$$C_X(t) = \frac{1}{L} \sum_{i=0}^{L-1} \text{Re} \left[\text{Tr}(\rho_0 \tilde{X}_i(t) \tilde{X}_i) - \text{Tr}(\rho_0 \tilde{X}_i(t)) \text{Tr}(\rho_0 \tilde{X}_i) \right]. \quad (16)$$

The subtraction removes the disconnected one-point contribution. Because ρ_0 need not commute with H_1 , this protocol is a thermal quench rather than a stationary equilibrium correlator. The time-dependent one-point function must therefore be retained in the subtraction. The code evaluates Eq. (16) in the eigenbasis of H_1 , allowing all requested times to be obtained from precomputed matrix contractions and phase factors. The local operator $\tilde{X}_i$ is a candidate probe of slow constrained dynamics; identifying the corresponding quasi-good quantum number or effective order parameter remains part of the Griffiths interpretation.

The current parameter set is $L = 16$, $g = -0.4$, $r = 0.2$, and $T = 1$. Times are sampled geometrically over $0.1 \leq t \leq 100$, which provides uniform resolution on a logarithmic time axis.

5.1 Power-law fit

The observed decay is modeled by

$$C_X(t) = A t^{-\alpha}. \quad (17)$$

For positive correlation values, Eq. (17) becomes the linear relation

$$\log C_X(t) = \log A - \alpha \log t. \quad (18)$$

Ordinary least-squares regression is applied directly to all correlation samples in the early-time interval

$$0.1 \leq t \leq t_{\text{fit}}, \quad t_{\text{fit}} = 50. \quad (19)$$

No peak detection or upper-envelope selection enters the fit. The fitted exponent is minus the regression slope, and the amplitude is the exponential of the intercept. The plot uses logarithmic axes and marks the fitted time interval with a translucent shaded region without an edge line. This regression measures whether a power law is consistent with the selected window. A model test must additionally fit a single exponential and a stretched exponential to the same untransformed data and compare their residual structure and window dependence.

Figure 5 shows the correlation data and the fitted power law. The regression gives $\alpha = 0.486$ for this parameter set. Oscillations remain around the fitted trend, but the dashed line captures the overall decrease across the shaded time window. Data beyond t_{fit} are displayed for comparison and do not enter the regression.

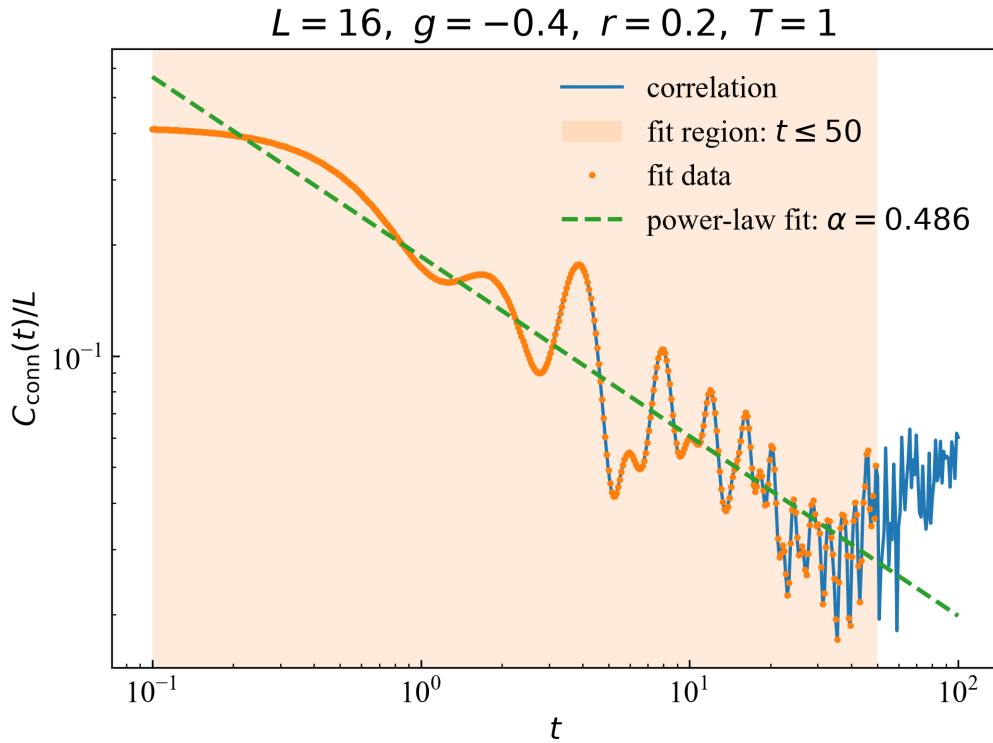


Figure 5: Connected local X -autocorrelation for $L = 16$, $g = -0.4$, $r = 0.2$, and $T = 1$ on logarithmic axes. Orange points inside the shaded region are the data used in the regression over $0.1 \leq t \leq 50$. The green dashed line is the fitted power law $C_X(t) = At^{-\alpha}$ with $\alpha = 0.486$; data at later times are excluded from the fit.

The approximately linear behavior in the log-log representation supports a power-law description of the finite-size autocorrelation over the selected window. Such slow relaxation is compatible with Griffiths phenomenology, in which a broad distribution of relaxation scales replaces a single decay time. The present model is clean and staggered rather than spatially random, and the fit

includes microscopic early times. The result is therefore an initial diagnostic, not a sufficient identification of a Griffiths phase. Its physical content must be tested against alternative decay models, shifted fit windows, additional temperatures and deformations, and increasing system size.

6 Tests for a disorder-free Griffiths regime

The numerical results above suggest a connection between a low-energy scarred subspace and slow relaxation. The following tests are needed to decide whether that connection defines a disorder-free Griffiths regime rather than a finite-size crossover.

1. **Joint dynamical and thermodynamic scaling.** The autocorrelation should favor an algebraic decay over a single exponential $Ae^{-t/\tau}$ and a stretched exponential $A \exp[-(t/\tau)^b]$ across a broad intermediate-time window. Independently, $C_V(T)$ should exhibit a compatible non-universal power law at low temperature. Both behaviors should persist over the same finite interval of g rather than at one finely tuned point.
2. **Separation between average and typical observables.** For a positive observable or response X , a broad distribution can be diagnosed by comparing

$$X_{\text{avg}} = \langle X \rangle, \quad X_{\text{typ}} = \exp(\langle \ln X \rangle). \quad (20)$$

A strong inequality $X_{\text{typ}} \ll X_{\text{avg}}$ would show that the mean is dominated by rare contributions. In this clean translation-invariant model, a site average cannot reproduce a disorder average. The relevant ensemble must first be defined, for example over eigenstates, symmetry sectors, phase-space regions, or controlled weak perturbations.

3. **Distribution of low-energy gaps.** If Δ samples the relevant low-energy or relaxation gaps, compare its mean with

$$\Delta_{\text{typ}} = \exp(\langle \ln \Delta \rangle). \quad (21)$$

The hierarchy $\Delta_{\text{typ}} \ll \langle \Delta \rangle$ would indicate a broad gap distribution. The ensemble used to construct this distribution must be specified before the comparison is meaningful.

4. **Negative checks.** Failure of a single exponential over a growing time window would distinguish the candidate regime from a conventional gapped paramagnet. Persistence of anomalous scaling over a finite parameter interval, together with finite-size analysis, would distinguish it from an isolated critical point.
5. **Microscopic origin of the slow sector.** The remaining conceptual task is to identify the approximate quantum number or phase-space structure that plays the role normally assigned to a locally ordered rare region. Establishing that structure would turn the observed analogy into a mechanism.

7 Summary

The deformed PXP chain provides a clean setting in which a scarred subspace is shifted toward low energy. Exact diagonalization identifies eigenstates with anomalously large $\mathbb{Z}_2$ overlap, symmetry-resolved spectra determine the canonical energy and heat capacity, and a thermal quench produces a connected local X -autocorrelation consistent with power-law decay over the fitted time

window. These results establish the finite-size spectral and dynamical ingredients of a possible connection between quantum many-body scars and Griffiths-like relaxation.

The stronger claim of a disorder-free Griffiths regime remains conditional. It requires consistent dynamical and heat-capacity exponents over a finite deformation window, a meaningful separation between average and typical observables, a broad distribution of gaps, and identification of the approximate structure responsible for rare slow events. Those tests define the next stage of the study and separate the proposed mechanism from ordinary crossover behavior or isolated criticality.